CMOR 504
Project Report

Project Report

Non-Isomorphic Local Complement Branch and Bound Scheme and its Applications to Graph Theory Fundamentals

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Abstract

Recent work in quantum computing focuses on understanding qubits in the lense of graph theory. One such operation includes the local complement. Despite its simple definiton, it can be become a redundant task if the graph is presumed to be dense and have local complements that are isomorphic to previous graphs. We're focused on constructing a recursive algorithm that finds all non-isomorphic graphs from the initial-state graph. Given the final listing of all the non-isomorphic graphs, we'll proceed to quantify the various properties of each graph including power domination, maximum stable set, cliques, matchings, and perfect matchings.

Introduction

Graph states in [5] connect qubits and both their computation and communication. One such focus comes from [1] on the use of local complements as a transformation on the original stated graph. Despite the local complement operation being straightforward, finding all possible combinations of such local complements can be challenging when the node count is large or the graph is densely connected. In fact, it can be redundant to end up seeing graphs that are isomorphic to ones already seen from preivous iterations. This results in a difficulty in collecting the information of each individual graph. These include the following:

(i) *Domination Number*

(ii) *Maximum Clique*

(iii) *Maximum Stable/Independent Set*

(iv) *Matchings*

(v) *Perfrect Matchings*

Hence, we propose a recursive algorithm that finds all non-isomorphic graphs to the original graph state. We'll proceed as follows:

(i) **Preliminary Material**: Reviewing the terminology and concepts to construct the recursive algorithm and the data to be collected on each graph.

(ii) **Methodology and Results**: Defining the recursive algorithm and presenting any results collected.

(iii) **Conclusions**: Summarize what's been done throughout this report.

(iv) **Limitations, Discussions, and Future Work**: Discuss the limitations and potential expansions to be performed on this report.

Preliminary Material

A **graph** G is an ordered pair, expressed as $G = (V, E)$ where V and E are the set of nodes and edges, respectively. Additionally, we assume that $|V|, |E| < \infty$. The **neighborhood** of a vertex $\mathbf{v} \in V$, denoted $N(v)$, is the set of all nodes $\mathbf{u} \in V$ such that $\mathbf{u}, \mathbf{v} \in E$.

The following definitions will be referenced from [4, 3, 2].

Definition

Let the graph $G = (V, E)$ be given.

(a) An **independent/stable set** is any collection of vertices $S \subseteq V$ such that no two vertices in S share an edge.

(b) The **maximum independent/stable set** problem finds the largest independent set $S \subseteq V$ to the graph G.

(c) A **clique** is a sest of vertices $K \subseteq V$ such that all pairs of vertiecs in K are adjacent to eachother.

(d) The **dominating set** is a subset $T \subseteq V$ such that each vertex G either belongs to G or is adjacent to some element of S.

(e) A **matching** is a set $M \subseteq E$ of pairwise disjoint edges.

(f) A **perfect matching** is a matching that covers each node in G.

We'll also be using the following definition from [1].

Definition

Let the graph $G = (V, E)$ be given.

Methodology and Results

Text on the methodology and results goes here . . . .

Conclusions

Text on the conclusions material goes here . . . .

Limitations, Discussions, and Future Work

Text on the limitations, discussions, and future work goes here . . . .

Bibliography

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